

Interactive Group Task: “Design, Evaluate & Visualise” Challenge

Duration: 1 Hour

Group Size: 3–5 Students

ACTIVITY OVERVIEW

Students will:

1. **Analyse an existing system** (interaction styles)
 2. **Redesign/improve the interaction**
 3. **Create a simple data visualisation concept**
 4. **Present and justify their design decisions**
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SCENARIO (Assign ONE per group)

Each group selects or is assigned one system:

- Mobile Banking App
 - E-learning Platform (LMS)
 - Online Shopping Website
 - Navigation App (e.g., maps)
 - Health/Fitness Tracking App
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TASK BREAKDOWN

PART 1: Identify Interaction Styles (15 minutes)

Task:

Analyse your assigned system and answer:

1. What interaction styles are used?
 - Command-based
 - Menu-based
 - Direct manipulation
 - Exploring/navigation
 - Others (voice, gesture, etc.)
 2. Give **specific examples**:
 - What actions do users perform?
 - How does the system respond?
 3. Identify:
 - Which interaction style is **most dominant**
 - Which one is **missing but could improve usability**
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PART 2: Usability Evaluation (15 minutes)

Task:

Evaluate the system using interaction characteristics:

- Feedback
- Visibility
- Consistency
- Learnability
- Efficiency
- Error prevention

Output:

Create a simple table:

Feature	Good/Bad	Explanation
Feedback	Good	Immediate response when clicking
Visibility	Poor	Important options hidden

Students must identify **at least 3 usability issues**

PART 3: Redesign the Interaction (15 minutes)

Task:

Propose improvements:

1. Redesign ONE key feature of the system
2. Explain:
 - What interaction style you are improving
 - How your design improves usability

Deliverable:

- Sketch (paper or digital)
 - OR bullet-point design explanation
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PART 4: Data Visualisation Task (10 minutes)

Scenario:

Your system collects data (e.g., user activity, sales, progress, routes, health stats)

Task:

Design a **simple interactive visualisation**:

1. What data will be shown?
2. What type of visualisation?
 - Bar chart
 - Line graph
 - Dashboard
 - Heatmap

3. What interactions will users perform?
 - Filter
 - Zoom
 - Click for details
4. Apply:
 - Overview → Zoom → Details principle

PART 5: Group Presentation (5 minutes per group)

Each group presents:

1. Interaction styles identified
2. Key usability issues
3. Improved design
4. Visualisation concept

EXPECTED LEARNING OUTCOMES

By the end of the activity, students should be able to:

- ✓ Identify and differentiate interaction styles
- ✓ Evaluate usability using key principles
- ✓ Apply interaction design improvements
- ✓ Design basic interactive visualisations
- ✓ Explain design decisions clearly

ASSESSMENT RUBRIC (Quick Marking Guide)

Criteria	Marks
Identification of interaction styles	20
Usability evaluation	20
Quality of redesign	20
Visualisation design	20
Presentation clarity	20
Total	100 Marks

EXTENSION

- Vote for the **best design**
- Discuss:
 - Which group had the most user-friendly solution?
 - Which visualisation was most effective?